

Vahan Stepanyan

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EDUCATION

Yerevan State University Mathematics and Mechanics	2014 – 2016
Yerevan State University Informatics and Applied Mathematics	2013 – 2014
Moscow Institute of Physics and Technology (State University) Control and Applied Mathematics	2011 – 2013
Yerevan State University Informatics and Applied Mathematics	2007 – 2011

EMPLOYMENT

Activeloop Staff R&D Engineer: <i>Research on speeding up vector similarity searches. Implemented and improved indexes for many data types, such text (IVF), quantized vector embeddings (CosSim) and matrix of embeddings (Col-Pali). Ensured the balance for adequate search accuracy and speed for Deeplake vector DBMS.</i>	2024 – 2026
Synopsys Inc. Staff R&D Engineer: Performance improvements on libraries and tools working with IC layouts. Embedded Python interpreter for the Windows version of the tool. Mentored and shared my experience with juniors.	2023 – 2024
Mobeus (Stealth Technology Startup) Senior Software Developer: <i>Dynamically linked libraries providing features accessible to Unity and Electron+Node systems on Windows. Participated in an audio driver development. Various shaders implementations and graphics programming via DirectX.</i>	2022 – 2023
TeamViewer Senior Software Developer: <i>Improvements for the main communication protocol. Features for IoT devices configuration and management application on Windows.</i>	2020 – 2022
VMware Inc. Software Developer: <i>System-programming part (not only) of a log collection agent with files (Linux + Windows) and Windows Event Log as sources for log entries.</i>	2013 – 2020
Institute for System Programming of Russian Academy of Sciences R&D Engineer: <i>GCC back-end optimization for better instructions emission for ARM based CPUs [Cortex family]. Features and bug-fixes on an instruction tracer and analyzer reverse engineering tool for Windows programs. Race condition detector for Java/Android applications.</i>	2011 – 2013
System Programming Lab at Yerevan State University R&D Engineer Intern: <i>Bash/coreutils based toolchain for compiler auto-tuner (optimal flags subset generator) performance measurement with constraints including input program compilation and running times, binary size and memory usage, involving various open source and in-house built c/c++ programs and libraries.</i>	2010 – 2011

COMPUTER LITERACY

- Programming languages: C/C++, Pascal/Delphi, x86/ARMv7, Bash/Batch, Python
- Libraries/Frameworks: WinAPI, POSIX, KOL, STL, Qt, Boost, POCO
- DBMS: SQLite, MS SQL, PostgreSQL
- Version Control: Git, Perforce, SVN
- Reverse Engineering: unpacking, disassembling, debugging, decompiling
- Optimization skills: complexity of algorithms, compiler-specific and platform-specific optimizations, profiling, low-level optimizations
- Packaging: MSI (Windows), RPM/DEB (Linux)
- Computer Architecture: PC, Embedded systems, microcontrollers
- Operating Systems: Linux (RedHat/Debian based), Windows

PROJECTS

<https://github.com/nop144d>

LANGUAGES

Armenian - native, Russian - fluent, English - good

PROFESSIONAL INTERESTS

system-programming, reverse engineering, demoscene, information security, hardware, malwares, packers/protectors, operating systems, compilers, optimizations, performance-critical and memory-critical programming, prime numbers, fractals, data compression, microcontrollers, robotics

NON-PROFESSIONAL INTERESTS

guitar, music theory